Mathematical Foundations of Political Methodology

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- Quadratic Form
- Positive Definiteness
- Multivariate Optimization

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Quadratic Form

Before we proceed our discussion to multivariate optimization, we introduce some concepts related to it. The first important concept is positive/negative definiteness.

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    - $ax^2 + bxy + cy^2$
    - $ax^2 + by^2 + cz^2 + dxy + exz + fyz$ (with coefficients a, b, c, d, e, f)

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    - $ax_2 + by$ is not a homogenous quadratic polynomial because one of the terms doesn't have degree 2

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where

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There is therefore a unique symmetric matrix $\mathbf{A}$ satisfying $q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$

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Takeaway point: A symmetric $\mathbf{A}$ is uniquely determined by the coefficients in $q(\mathbf{x})$

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- Notice that we can rewrite it as: $q(\mathbf{x}) = x_1^2 + (x_2 - x_3)^2 \geq 0$

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Example: $q(x, y) = xy$

- $q(1, 1) > 0$
- $q(1, -1) < 0$

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- $\mathbf{A}$ is NSD iff $a, d \leq$ and $\det \mathbf{A} \geq 0$

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- Policy $\boldsymbol{x} \in \mathbb{R}^n$ that maximizes $U(\boldsymbol{x})$
- Weights $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_K)$ such that a weighted average of forecasts $\boldsymbol{f} = (f_1, f_2, \ldots, f_k)$ have minimum loss

$$\min_{\boldsymbol{\pi}} \quad = \quad -(\sum_{j=1}^{K} \pi_j f_j - y)^2$$

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We'll describe analytic and computational approaches to optimization

Multivariate Optimization

Definition

Let $\boldsymbol{x} \in \mathbb{R}^n$ and let $\delta > 0$. Define a neighborhood of $\boldsymbol{x}$, $B(\boldsymbol{x}, \delta)$, as the set of points such that,

$$B(\boldsymbol{x}, \delta) \quad = \quad \{\boldsymbol{y} \in \mathbb{R}^n : ||\boldsymbol{x} - \boldsymbol{y}|| < \delta\}$$

Multivariate Optimization

Definition

Suppose $f : X \rightarrow R$ *with* $X \subset R^n$. *A vector* $\boldsymbol{x}^* \in X$ *is a global maximum if , for all other* $\boldsymbol{x} \in X$

$$f(\boldsymbol{x}^*) \;\; > \;\; f(\boldsymbol{x})$$

A vector $\boldsymbol{x}^{local}$ *is a local maximum if there is a neighborhood around* $\boldsymbol{x}^{local}$, $Q \subset X$ *such that, for all* $x \in Q$,

$$f(\boldsymbol{x}^{local}) \;\; > \;\; f(\boldsymbol{x})$$

Gradient

Definition

Suppose $f : X \to \mathbb{R}^n$ with $X \subset \mathbb{R}^1$ is a differentiable function. Define the gradient vector of f at $\boldsymbol{x}_0$, $\nabla f(\boldsymbol{x}_0)$ as,

$$\nabla f(\boldsymbol{x}_0) \quad = \quad \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_3}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_n}\right)$$

Gradient First Order Condition

Theorem

Suppose $f : X \to R^1$, $X \subset R^n$. *Suppose* $\boldsymbol{a} \in X$ *is a local extremum. Then,*

$$\begin{aligned} \nabla f(\boldsymbol{a}) &= \mathbf{0} \\ &= (0, 0, \ldots, 0) \end{aligned}$$

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- Critical Values:
    1) Maximum
    2) Minimum
    3) Saddle point
- Second Derivative Test!

Second Order Conditions: Hessian

Definition

Suppose $f : X \to R^1$, $X \subset R^n$, with f a twice differentiable function. We will define the Hessian matrix as the matrix of second derivatives at $\boldsymbol{x}^* \in X$,

$$\boldsymbol{H}(f)(\boldsymbol{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\boldsymbol{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\boldsymbol{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\boldsymbol{x}^*) \end{pmatrix}$$

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General test $\rightsquigarrow$ Two Dimensional Test $\rightsquigarrow$ Example

Second Derivative Test

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Theorem

Two Dimensional Second Derivative Test. Suppose $f : X \to \mathbb{R}$ with $X \subset \mathbb{R}^2$ and f twice differentiable. Write the Hessian of f at a critical value $\boldsymbol{a}$,

$$\boldsymbol{H}(f)(\boldsymbol{a}) = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

Then, we can conduct the second derivative test as:

- $AC - B^2 > 0$ *and* $A > 0 \rightsquigarrow$ *positive definite* $\rightsquigarrow$ $\boldsymbol{a}$ *is a local minimum*
- $AC - B^2 > 0$ *and* $A < 0 \rightsquigarrow$ *negative definite* $\rightsquigarrow$ $\boldsymbol{a}$ *is a local maximum*
- $AC - B^2 < 0 \rightsquigarrow$ *indefinite* $\rightsquigarrow$ *saddle point*
- $AC - B^2 = 0$ *inconclusive*

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5) Boundary values? (if relevant)

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$$\begin{aligned} \nabla f(\boldsymbol{x}) &= (6x_1 + 12, 8x_2 + 32) \\ \mathbf{0} &= (6x_1^* + 12, 8x_2^* + 32) \end{aligned}$$

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We now solve the system of equations to yield $x_1^* = -2$ and $x_2^* = -4$

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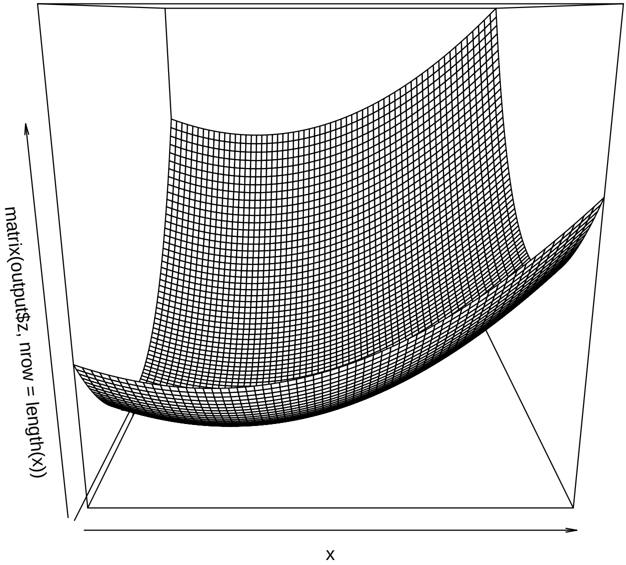
$\det(\mathbf{H}(f)(\boldsymbol{x}^*)) = 48$ and $6 > 0$ so $\mathbf{H}(f)(\boldsymbol{x}^*)$ is positive definite. local minimum


matrix(output$z, nrow = length(x))
x

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$$\begin{aligned} -2(x_1^* - \mu_1) &= 0 \\ -2(x_2^* - \mu_2) &= 0 \end{aligned}$$

Example 2: Two Dimensional Ideal Points

Suppose legislators are considering legislation $\boldsymbol{x} \in \mathbb{R}^2$. And suppose legislator i has utility function $U_i : \mathbb{R}^2 \to \mathbb{R}$,

$$U(\boldsymbol{x})_i = -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2$$

What is legislator i's optimal policy?
$\nabla f(\boldsymbol{x}) = (-2(x_1 - \mu_1), -2(x_2 - \mu_2))$
$\nabla f(\boldsymbol{x}) = \mathbf{0}$

$$\begin{aligned} -2(x_1^* - \mu_1) &= 0 \\ -2(x_2^* - \mu_2) &= 0 \end{aligned}$$

Solving yields $x_1^* = \mu_1$ and $x_2^* = \mu_2$.

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So, $-2 * -2 - 0 = 4 > 0$ and $-2 < 0 \rightsquigarrow$ negative definite, maximum
$\boldsymbol{\mu} = (\mu_1, \mu_2)$ are legislator i's two dimensional ideal point.